Adaptive Multi-Agent Learning Support through Deterministic Learner-State-Based Agent Selection

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***Abstract*—Multi-agent systems can provide different forms of learning support, but activating several agents for every learner interaction can increase resource use and may result in unnecessary processing. This paper presents a lightweight multi-agent framework for personalised learning support in which a deterministic, rule-based decision engine uses the learner's current state to select the next instructional action and the agents required for it. The contribution is an efficient orchestration mechanism for educational agents rather than a learned adaptive policy. The framework uses specialised agents for learner analysis, planning, tutoring and assessment, together with a deterministic decision engine for selecting the next instructional action. Learner mastery is updated after each assessed response, and repeated misconceptions can trigger remediation. Four instructional actions are considered: explain, practice, remediate and challenge. The framework is evaluated using a controlled benchmark of information technology learning questions and scripted learner profiles with different mastery and error patterns. It is compared with a single-agent baseline and a fixed multi-agent system using action-selection accuracy, feedback quality and resource use. An ablation study examines the effect of learner-state information and adaptive agent selection. The adaptive system achieved 67.86% action-selection accuracy, compared with 47.62% for the single-agent baseline and 55.95% for the fixed multi-agent system. Compared with the fixed multi-agent system, it used 735.6 fewer tokens and 1.88 s less per interaction, while the difference in action-selection accuracy was not significant ($p = 0.0525$). Removing learner-state information reduced action-selection accuracy to 39.29%, while removing dynamic Tutor/Remediation selection did not change action-selection accuracy.**

***Index Terms*—Adaptive learning, multi-agent systems, personalised learning, learner modelling, agent selection**

I. Introduction

Personalised learning aims to adapt learning activities to individual learner needs, knowledge and progress. However, learner knowledge can change as learning continues, so a learning path that is suitable at one stage may not remain appropriate later. Earlier work by Hans, Chakraverty and Bindal treated learning ability as a changing learner characteristic and used learner performance to modify subsequent learning paths [1]. This provides a basis for maintaining learner state and using updated performance information when making subsequent learning decisions.

Adaptive e-learning research has considered different approaches to representing learner state and using it for content adaptation. A recent review of learner models found that knowledge level and other learning-related characteristics are frequently used to support adaptation, although learner states are represented and applied in different ways across systems [2]. For an adaptive learning system, maintaining such information is useful when it can be translated into an appropriate next learning activity.

Multi-agent systems offer another way to organise learning support by assigning different tasks to specialised agents. For example, separate agents can be used for analysing learner responses, planning activities, providing tutoring support and assessing subsequent responses. Recent research on multi-agent learning environments has also reported differences in learner interaction patterns associated with prior knowledge [3]. An important practical issue is whether all agents need to be involved in every interaction. When a learner has demonstrated sufficient understanding, activating the complete set of agents may provide little additional benefit. In contrast, repeated errors or misconceptions may require additional analysis and targeted support.

In this study, a lightweight multi-agent framework is developed in which agent activation depends on the current learner state. The framework uses specialised agents for learner analysis, planning, tutoring and assessment, while a deterministic decision engine selects the next instructional action. Learner mastery is updated after each assessed response, and repeated misconceptions can trigger targeted remedial activities. This allows the system to vary the amount and type of processing according to the learner's observed performance.

The proposed framework is compared with a single-agent baseline and a fixed multi-agent configuration in which all agents are activated for every interaction. The experiments use information technology questions and scripted learner profiles representing different performance patterns. The evaluation considers learning-support quality together with agent calls, token usage and response latency. An ablation study examines the effect of learner-state information and adaptive agent selection. The contribution is therefore an efficient orchestration mechanism for educational agents under a fixed, pre-specified routing policy, rather than a learned policy or a demonstrated improvement in learning.

II. Related Work

Personalised and adaptive learning systems commonly use learner models to represent knowledge, performance and other learner characteristics. Wang et al. reviewed 57 studies on adaptive e-learning and found that knowledge level and other learning-related characteristics were frequently used for adaptation. They also reported differences in how learner models are developed and how learner information is used to guide adaptation decisions [2]. Böck et al. similarly examined learner model design and found that learning systems differ in the components and structures used to represent learners [4]. These studies support the use of an explicit learner state when learner information needs to be updated during successive interactions.

Personalised adaptive learning has also been studied in higher education. Du Plooy et al. reviewed 69 studies and examined the characteristics and reported effects of personalised adaptive learning on academic performance and engagement [5]. Tan et al. reviewed adaptive learning platforms, covering their architectures, adaptation methods and evaluation approaches [6]. Hariyanto et al. examined recent applications of intelligent methods in adaptive education [7]. The studies show that adaptive systems can use different learner information and adaptation mechanisms, depending on the learning setting and system design.

Recent work has extended personalised learning to systems with multiple specialised agents. Wang et al. proposed GenMentor, a multi-agent framework that uses learner goals and skill gaps to generate personalised learning paths [8]. Hedi et al. developed a personalised adaptive learning framework with separate agents for tutoring, practical activities and teaching, supported by a decision layer [9]. Sun and Tai presented MultiTutor, where multiple language-model-based agents provide different forms of student support, including explanations and practice activities [10]. Hao et al. studied student interactions with multiple instructional agents in a university course and reported different interaction patterns for learners with different levels of prior knowledge [3].

Agent-based learner simulation has also been used to evaluate educational systems. Gao et al. proposed Agent4Edu, which uses generative agents to produce learner responses based on learner profiles and interaction history [11]. Such controlled learner scenarios can be useful when evaluating adaptive behaviour without presenting simulated learners as real student participants.

Existing studies have addressed different aspects of adaptive multi-agent learning, including learner modelling, personalised adaptation and the use of specialised instructional agents. As shown in Table I, these elements have been combined in different ways across existing systems. In this work, we focus on a more specific question: whether specialised agents can be activated only when they are needed, based on a small set of learner-state information and fixed decision rules. We therefore evaluate learning-support quality together with the resource use associated with agent activation.

III. Methodology

A. System Design

The proposed system uses four specialised agents: Learner Analysis, Planning, Tutor/Remediation, and Assessment. A deterministic decision engine selects the next learning action and determines the agents required for that action. The learner state contains the learner identifier, topic, mastery score, difficulty level, recent accuracy, error count, last action, misconception and interaction count.

The overall flow of the system is shown in Fig. 1.

The Learner Analysis Agent examines the question, learner response and previous interaction information. The Assessment Agent evaluates the response and provides the response score used to update learner mastery. The Tutor/Remediation Agent generates the corresponding learning support when required. The Planning Agent is used in the fixed multi-agent configuration, whereas the adaptive configuration uses the deterministic decision engine to determine the required action and agent participation.

The decision engine uses the updated learner state to select one of four actions: EXPLAIN, PRACTICE, REMEDIATE, or CHALLENGE. In System C, the same learner state always produces the same action.

B. Learner State Update and Action Selection

After an assessed response, the response score R_t is normalised to the range $[0, 1]$ and used to update the learner's mastery:

$$M_{t+1} = 0.7M_t + 0.3R_t. \tag{1}$$

Here, M_t is the mastery before the response and M_{t+1} is the updated mastery. The same update rule is used throughout the experiments. The 0.7 weight limits the effect of a single response; from the initial value of 0.50, three consecutive full-score responses are needed to reach the CHALLENGE band. The updated mastery is then used to select the next instructional action.

The action is selected using the following fixed thresholds:

$$A_{t+1} = \begin{cases} \text{CHALLENGE}, & M_{t+1} \geq 0.80, \\ \text{PRACTICE}, & 0.60 \leq M_{t+1} < 0.80, \\ \text{EXPLAIN}, & 0.40 \leq M_{t+1} < 0.60, \\ \text{REMEDIATE}, & M_{t+1} < 0.40. \end{cases} \tag{2}$$

The bands provide graded support, from remediation of foundational gaps below 0.40 to more demanding tasks at 0.80 and above. The thresholds and weights were fixed before development and were not tuned on either question set. The repeated-misconception check is applied before the mastery thresholds. When the same misconception occurs twice consecutively on the same topic, REMEDIATE is selected regardless of the mastery score. The misconception streak is reset after remediation, a correct response, or a different misconception.

TABLE I
COMPARISON OF RECENT WORK ON PERSONALISED AND MULTI-AGENT LEARNING SYSTEMS.

Study	Learner state	Personalised adaptation	Multiple agents	Selective activation	Deterministic routing	Resource Evaluation
Wang et al. [2]	✓	✓	–	–	–	–
Böck et al. [4]	✓	✓	–	–	–	–
du Plooy et al. [5]	✓	✓	–	–	–	–
Tan et al. [6]	✓	✓	–	–	–	–
Wang et al. [8]	✓	✓	✓	–	–	–
Hedi et al. [9]	✓	✓	✓	–	–	–
Sun and Tai [10]	✓	✓	✓	–	–	–
Hao et al. [3]	–	✓	✓	–	–	–
Gao et al. [11]	✓	✓	✓	–	–	–
Hariyanto et al. [7]	✓	✓	–	–	–	–
Proposed framework	✓	✓	✓	✓	✓	✓

Note: A checkmark indicates that the corresponding feature is explicitly described in the cited work; "–" indicates that the feature was not identified.

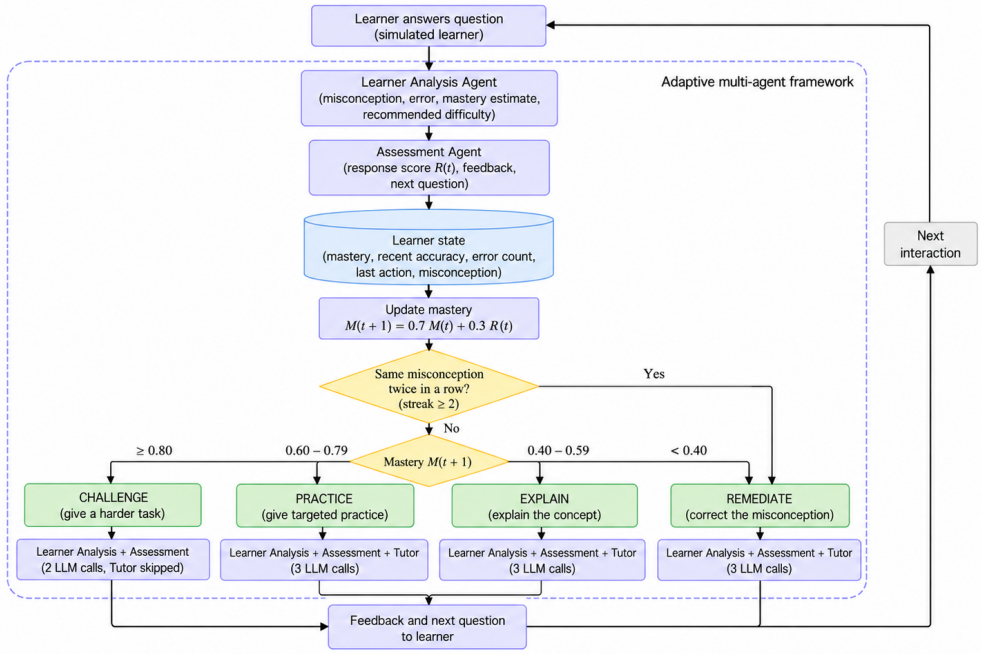


Fig. 1. Overview of the proposed methodology.

For a new learner, the initial mastery score is set to 0.50. Recent accuracy and misconception are initially empty, the error count is zero, and no previous action is recorded.

The selected action also determines the agents used in the adaptive configuration. The Tutor/Remediation Agent is skipped when CHALLENGE is selected and is used for PRACTICE, EXPLAIN and REMEDIATE.

C. System Configurations

Three configurations are used in the experiments. System A is the single-agent baseline. One LLM receives the learner information and produces the evaluation, feedback and next activity. System B is the fixed multi-agent baseline, in which the Learner Analysis, Planning, Tutor/Remediation, and Assessment Agents are activated for every interaction. System C is the proposed adaptive system, where the decision engine activates only the agents required for the selected action. Systems A and B use the LLM to select the action, while System C selects the action using the decision engine. The comparison therefore varies both the agent configuration and whether the routing policy is executed in code or by the LLM.

In System C, the Planning Agent is not called as a standard part of every interaction. The Tutor/Remediation Agent is also skipped when CHALLENGE is selected.

Gemma 3 4B was used for all three systems and was run on an AMD Ryzen 7 7435HS, RTX 4050 Laptop GPU and 16 GB RAM. The same model version, temperature, questions, initial learner state and scoring procedure are used for all three systems. Prompts are fixed before the final evaluation. The locked evaluation questions are not used for prompt adjustment or system tuning.

Action-selection accuracy is the proportion of interactions

in which the selected action matches the action obtained by applying the routing rules to the reference scores of each scripted learner. Because System C executes these rules, its accuracy mainly reflects whether its assessment produces the learner state the rules require, not whether the rules are pedagogically optimal. Rule consistency is agreement with the rules applied to the system's own assessed score.

D. Dataset and Learner Profiles

The benchmark contains 40 information technology questions, with 10 questions each from Database, Operating Systems, Programming and Cybersecurity. Twenty questions are used during development and 20 are kept as the locked evaluation set.

Each question includes the topic, concept, difficulty, expected answer and scoring guidance. Learner behaviour is represented by scripted profiles covering strong, moderate and weak mastery conditions. Repeated-error cases are included. We have used scripted profiles to provide controlled learner conditions to evaluate the routing rules and represent system test cases rather than real student participants.

The locked evaluation set produces 84 evaluated interactions for each system configuration from 44 learners with shared state. The same evaluation cases are used across the three configurations to support paired comparison. The set size was limited by the 20 locked questions and local model run time, which reduces statistical power, as seen in the C versus B comparison.

E. Evaluation

Action-selection results are compared using the exact McNemar test because the same evaluation cases are used across configurations. Feedback quality was evaluated using an automated LLM judge, Gemma 3 12B, with the judge_v3 prompt, and differences are summarised using a 95% bootstrap interval. The judge was run on an AMD Ryzen 9 9950X (16 cores), RTX 4070 SUPER and 64 GB DDR5. The resource-use measures are LLM calls, token usage and latency.

For the adaptive behaviour test, each selected action is compared with the reference action for that interaction.

Two ablation settings are used. In the first, learner-state information is removed from the adaptive system. In the second, dynamic Tutor/Remediation selection is removed. These settings are compared with the full adaptive system.

Since the interactions are grouped within 44 learners, learner-cluster intervals are also reported. The learner-cluster interval for the difference between System C and System A was +0.061 to +0.341, while the interval for the difference between System C and System B was +0.014 to +0.220.

F. Data Logging

For each interaction, the system records the learner state, response correctness, selected action, activated agents, mastery values, error information, token usage, latency and evaluation scores. Raw logs are retained separately from the processed results.

IV. Results and Discussion

A. Comparison of System Configurations

Table II gives the results for the three system configurations. Action-selection accuracy was 47.62% for System A, 55.95% for System B and 67.86% for System C. The difference between System C and System A was 20.24 percentage points, while the difference between System C and System B was 11.91 percentage points.

The higher accuracy of System C suggests that executing the routing policy in code is more reliable than asking the LLM to apply it. The difference was larger against the single-agent configuration than against the fixed multi-agent configuration. However, the difference between System C and System B should be interpreted with care because the exact McNemar test gave $p = 0.0525$.

Feedback quality followed a different pattern. The mean scores were 4.53 for System A, 4.35 for System B and 4.37 for System C. The adaptive configuration therefore did not give the highest feedback quality. The higher action-selection accuracy was not accompanied by a higher feedback quality score.

The number of LLM calls was 1.00 for System A, 4.00 for System B and 2.88 for System C. Token usage was 1066.5, 2944.0 and 2208.4 tokens, respectively. Mean latency was 3.84 s, 10.42 s and 8.54 s. Compared with System B, System C used 735.6 fewer tokens and required 1.88 s less per interaction.

These results show the main difference between the three configurations. The single-agent configuration used the fewest calls, tokens and time. The fixed multi-agent configuration used the most. The adaptive configuration was between the two in resource use and had the highest observed action-selection accuracy.

Figure 2 shows the token usage and latency breakdown for the three system configurations and the two adaptive ablations.

TABLE II
Comparison of the three system configurations.

System	Action Accuracy	Feedback Quality	LLM Calls	Tokens	Latency (s)
A – Single	47.62%	4.53	1.00	1066.5	3.84
B – Fixed	55.95%	4.35	4.00	2944.0	10.42
C – Adaptive	67.86%	4.37	2.88	2208.4	8.54

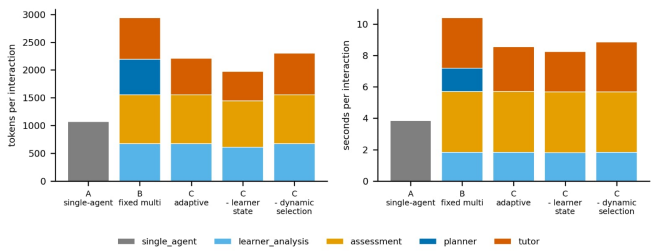


Fig. 2. Token usage and latency breakdown by system configuration and agent type.

B. Action Selection

The difference between Systems C and A was significant (exact McNemar $p = 0.0137$), while the difference between Systems C and B was not ($p = 0.0525$).

Removing learner-state information reduced the action-selection accuracy of System C from 67.86% to 39.29%, a decrease of 28.57 percentage points. The exact McNemar test gave $p = 0.0007$. The reduction is consistent with the use of learner-state information, particularly the updated mastery and misconception-related information used by the routing process.

Figure 3 shows the confusion matrices for Systems A, B and C. The system correctly selected 10 of 19 CHALLENGE cases, 12 of 22 PRACTICE cases, 18 of 23 EXPLAIN cases and 17 of 20 REMEDIATE cases. Most errors were incorrect selections of REMEDIATE (15 of 27), and 13 of these came from the repeated-misconception override, which relies on the Learner Analysis Agent reusing the same free-text misconception label. The system correctly identified most EXPLAIN and REMEDIATE cases.

Repeated misconceptions also affected action selection. The repeated-misconception override was triggered 18 times. In 13 cases, the override replaced an action that was already correct, while 5 cases matched the expected action. The override follows the rule that repeated occurrence of the same misconception on the same topic selects REMEDIATE.

System B correctly selected all 20 REMEDIATE cases, but its lower accuracy was mainly due to over-selecting REMEDIATE. System C showed its main gain in EXPLAIN, while CHALLENGE remained the lowest, with 10 of 19 cases selected correctly, compared with 12 for System A and 11 for System B.

C. Ablation Results

Table III shows the results of the two ablation settings. Removing the learner-state information reduced action-selection accuracy from 67.86% to 39.29%. The difference was 28.57 percentage points, with an exact McNemar test giving $p = 0.0007$. Rule consistency also decreased from 1.00 to 0.417. Feedback quality decreased from 4.37 to 4.26. The correctness score decreased from 3.40 to 3.14, while completeness changed from 5.00 to 4.98 and relevance from 5.00 to 4.86.

For the second ablation, dynamic Tutor/Remediation selection was removed. Action-selection accuracy remained at 67.86%, with an exact McNemar test giving $p = 1.0000$. Rule consistency also remained at 1.00. Feedback quality changed from 4.37 to 4.38. The correctness score increased from 3.40 to 3.43, while completeness remained at 5.00 and relevance remained at 5.00.

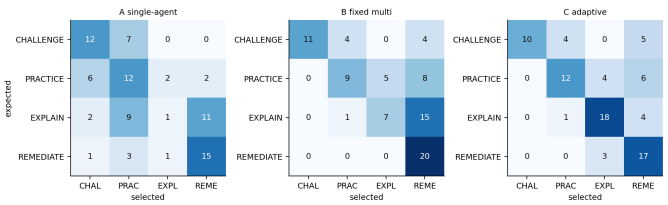


Fig. 3. Confusion matrices for action selection in Systems A, B and C.

TABLE III
ABLATION RESULTS FOR THE ADAPTIVE SYSTEM.

Configuration	Action Selection Accuracy	Feedback Quality	LLM Calls	Tokens	Latency (s)
C – Adaptive	67.86%	4.37	2.88	2208	8.54
C – Learner State Removed	39.29%	4.26	2.74	1976	8.25
C – Dynamic Selection Removed	67.86%	4.38	3.00	2300	8.87

TABLE IV
SUMMARY OF LEARNER TRAJECTORY CASES.

Profile	Observed pattern
Recovering	Mastery changes across the interactions.
High-mastery	Starts at CHALLENGE and transitions to REMEDIATE on Turn 2 due to the repeated-misconception override, despite mastery remaining above 0.80.
Cold-start	The cold-start case remains at lower-level support as mastery decreases.
Volatile	Actions change as the response scores vary across interactions.
Declining	Lower response scores lead to EXPLAIN and REMEDIATE.
Repeated-misconception	Repeated errors can trigger REMEDIATE according to the routing rule.

Accuracy is unchanged because the action is selected before the Tutor/Remediation Agent is called and its output does not affect mastery; removing routing only adds a Tutor call on the 10 of 84 CHALLENGE turns. Thus, the two ablations showed different effects. Removing learner-state information reduced action-selection accuracy, whereas removing dynamic Tutor/Remediation selection left action-selection accuracy unchanged but increased resource use.

D. Learner Trajectories

Figure 4 shows six learner trajectories used in the evaluation: recovering, high-mastery, cold-start, volatile, declining and repeated-misconception. The trajectories show how the mastery value and selected action change over successive interactions.

The recovering case shows changes in mastery across the interactions. The high-mastery case (S10-DB) starts at CHALLENGE and transitions to REMEDIATE on Turn 2 because the repeated-misconception override is triggered, even though mastery remains above 0.80. The cold-start case remains at lower-level support as mastery decreases. The volatile case changes between actions as the response scores change, while the declining case moves towards EXPLAIN and REMEDIATE as performance decreases.

The trajectories also contain cases where the selected action differs from the expected action. These cases are marked with "!" in the figure.

E. Feedback Quality and Error Propagation

The feedback-quality results were different from the action-selection results. System A had a mean quality score of 4.53, while System C had a score of 4.37. The paired difference was -0.16, with a 95% bootstrap interval of $[-0.26, -0.06]$.

The assessment step can also affect later actions. In one evaluation case, the reference response score was 0.85, while the system assigned a score of 0.40. The lower score changed

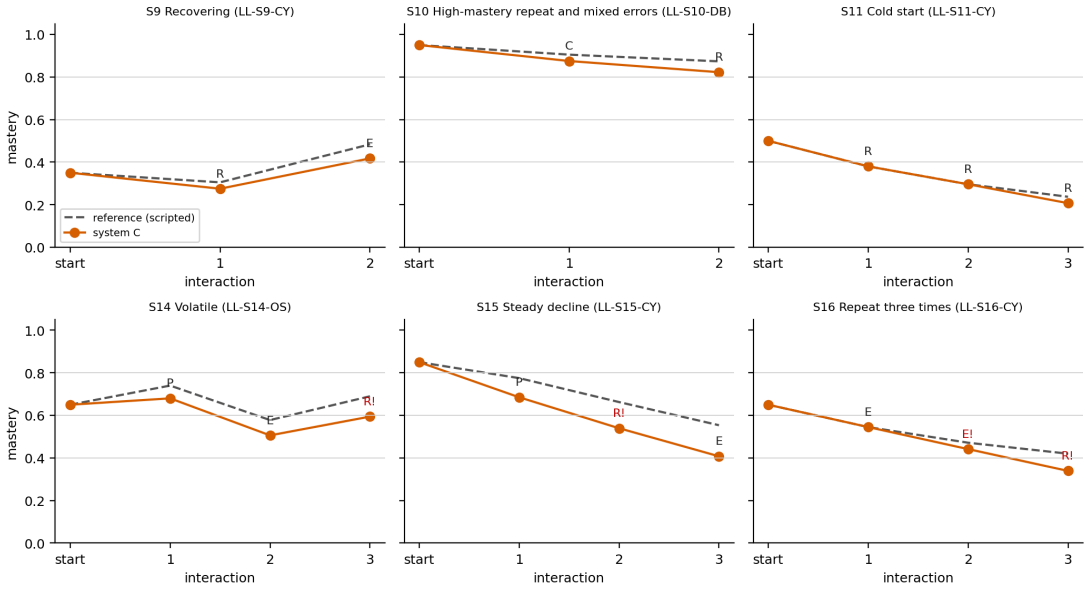


Fig. 4. Example learner trajectories showing changes in mastery and selected actions across successive interactions.

the learner mastery value and affected the following action. Since the decision engine uses the assessed response to update mastery, an assessment error can therefore affect more than the current interaction.

V. Conclusion

The experiments compared three configurations for adaptive learning support: a single-agent system, a fixed multi-agent system and the proposed adaptive system. The adaptive system selected the next action from the learner state and used the selected action to decide which agents were needed.

System C reached 67.86% action-selection accuracy, compared with 47.62% for System A and 55.95% for System B. Removing learner state reduced it to 39.29%, while removing dynamic selection only increased resource use.

The results were not the same for all measures. System A had the highest feedback quality score, while System C used fewer calls, tokens and time than System B. System C therefore did not give the highest value for every measure. The assessment-error case also showed that an incorrect response score can change the learner state and affect the following action.

The experiments used scripted learner profiles. The results are therefore limited to the system behaviour tested in these cases. They do not show whether the approach improves learning for real students. An important next step is evaluation with real learners using pre- and post-tests, retention, misconception resolution and student and instructor feedback, to test whether the resource savings bring educational benefit.

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